4.6 Integrals With No Formula

WORKSHEET

The question

(i) Given that

$$\int \frac{x^3-2}{(x+1)^2}\,e^x\,\mathrm{d}x = \frac{\mathrm{P}(x)}{\mathrm{Q}(x)}\,e^x + \text{constant},$$

where $\mathrm{P}(x)$ and $\mathrm{Q}(x)$ are polynomials, show that $\mathrm{Q}(x)$ has a factor of $x+1$. Show also that the degree of $\mathrm{P}(x)$ is exactly one more than the degree of $\mathrm{Q}(x)$, and find $\mathrm{P}(x)$ in the case $\mathrm{Q}(x) = x+1$.

(ii) Show that there are no polynomials $\mathrm{P}(x)$ and $\mathrm{Q}(x)$ such that

$$\int \frac{1}{x+1}\,e^x\,\mathrm{d}x = \frac{\mathrm{P}(x)}{\mathrm{Q}(x)}\,e^x + \text{constant}.$$

You need consider only the case when $\mathrm{P}(x)$ and $\mathrm{Q}(x)$ have no common factors.

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself If $\int f(x)e^x\,dx = \frac{P}{Q}e^x + C$, what equation must P and Q satisfy? (Differentiate the right-hand side.)

STAGE 1 The factor $x+1$

Ask yourself Put $x=-1$ in (†). What must happen?

STAGE 2 Degrees

Ask yourself In $P'Q - PQ' + PQ$, which term has the highest degree? Can it be cancelled by the others?

STAGE 4 An impossibility

Pause and try Substitute $Q = (x+1)^k R$, cancel as many factors of $(x+1)$ as you can, and then put $x=-1$. What do you learn about $P(-1)$?

Try this next

1 **Two that work.** Show that $\int \frac{xe^x}{(x+1)^2}\,dx = \frac{e^x}{x+1} + C$ and $\int \frac{(x^2+1)e^x}{(x+1)^2}\,dx = \frac{x-1}{x+1}e^x + C$.

2 **The Gaussian.** Show that there is no polynomial P with $\int e^{-x^2}\,dx = P(x)e^{-x^2} + C$. (*Hint:* differentiate and compare degrees.)

3 **Another pole.** Adapt the argument of part (ii) to show that $\int \frac{e^x}{x}\,dx$ is not of the form $\frac{P(x)}{Q(x)}e^x + C$.

4 **When is it possible?** Suppose f is a polynomial and $\int \frac{f(x)}{(x+1)^2}e^x\,dx = \frac{P(x)}{x+1}e^x + C$ for some polynomial P. Show that $(x+1)P' + xP = f$, and by evaluating this and its derivative at $x=-1$, show that $f(-1) + f'(-1) = 0$. Check that this condition holds for $f(x) = x^3 - 2$ and for the examples in exercise 1. HARDERCan you show that the condition is also sufficient?

Further reading

- **Brian Conrad, “Impossibility theorems for elementary integration”**. Freely available lecture notes explaining Liouville’s theorem with examples.
- **Elena Anne Marchisotto & Gholam-Ali Zakeri, “An invitation to integration in finite terms”,** *The College Mathematics Journal* 25 (1994). A gentle introduction to Liouville’s theory.
- **Manuel Bronstein, *Symbolic Integration I*** (Springer). How computer algebra systems actually integrate. Advanced.

The full guided solution, and what the question leads to at university, is in Section 4.6:
https://beyondstep.keithmcnulty.org/ch04/no-formula.html